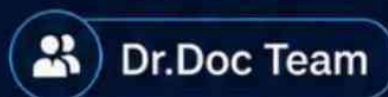




Electro cardiogram/**ECG**

— Prepared by —



P
Atrial
Depolarization

Q
Onset of
Ventricular
Depolarization

R
Ventricular
Depolarization

S
Ventricular
Repolarization
Begins

T
Ventricular
Repolarization



Essential for
Clinical Practice



Accurate
Interpretation



Conceptual
Clarity



Exam
Focused



Better Diagnosis
Better Care

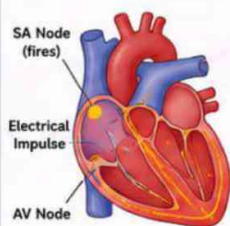
“ Understand the rhythm,
Save a life. ”

HOW IS THE ECG GENERATED?

An electrocardiogram (ECG) is a recording of the electrical activity of the heart as it is conducted to the surface of the body.

1 Electrical Activity in the Heart

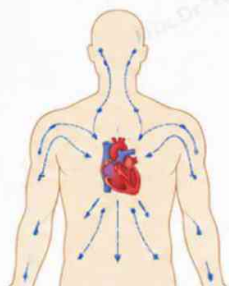
When the heart cells depolarize and repolarize, tiny electrical currents are generated.



Source of electrical activity: SA node

2 Spread of Electrical Currents

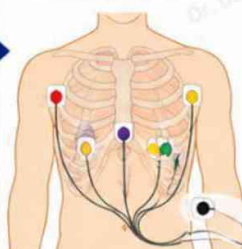
These electrical currents spread through the heart muscle, body fluids and tissues.



Body fluids act as conductors of electricity

3 Detection by Electrodes

Surface electrodes placed on the skin detect the voltage differences caused by these electrical currents.



Electrodes act as voltage sensors

4 Transmission to the ECG Machine

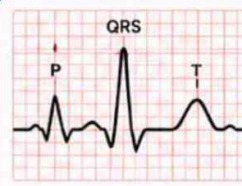
The electrodes send the detected signals as electrical impulses to the ECG machine.



Signals are amplified and processed

5 Display as ECG Waveform

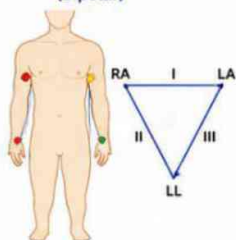
The machine interprets these signals and displays them as waves (P, QRS, T) on a graph – the ECG.



This is the Electrocardiogram

TYPES OF ECG LEADS

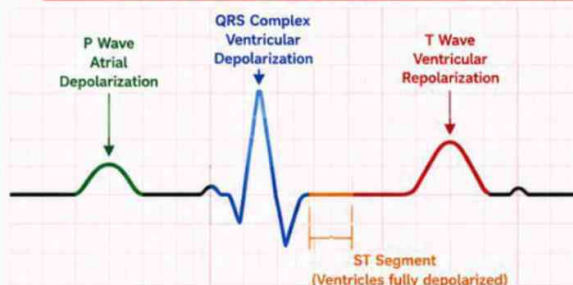
Limb Leads (Bipolar)



Chest Leads (Unipolar)



NORMAL ECG WAVEFORM & WHAT IT REPRESENTS



KEY POINTS

- ECG does not measure contraction.
- It records electrical events.
- Each wave corresponds to a specific electrical activity in the heart.
- It is a non-invasive, painless and essential diagnostic tool.

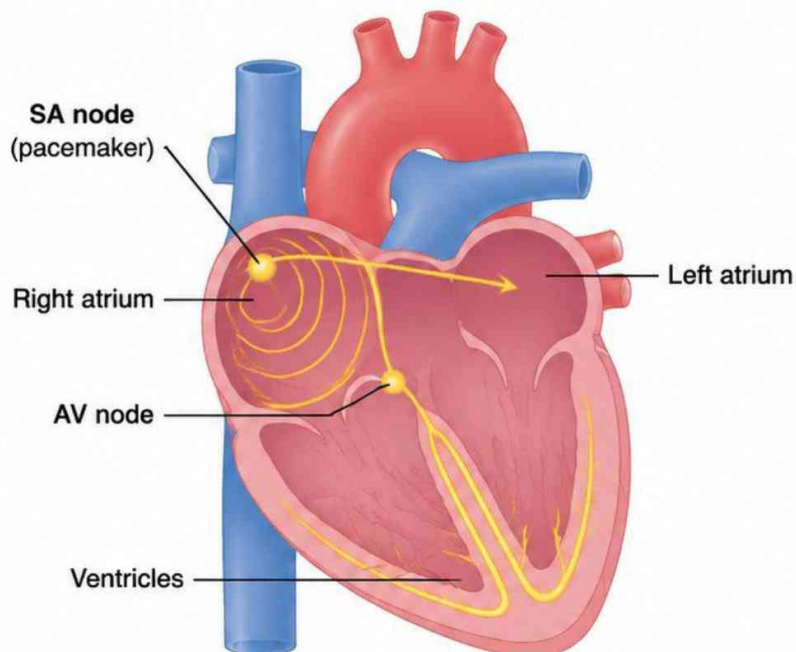


In short: Heart's Electrical Activity → Body Tissues → Electrodes → ECG Machine → ECG Waveform

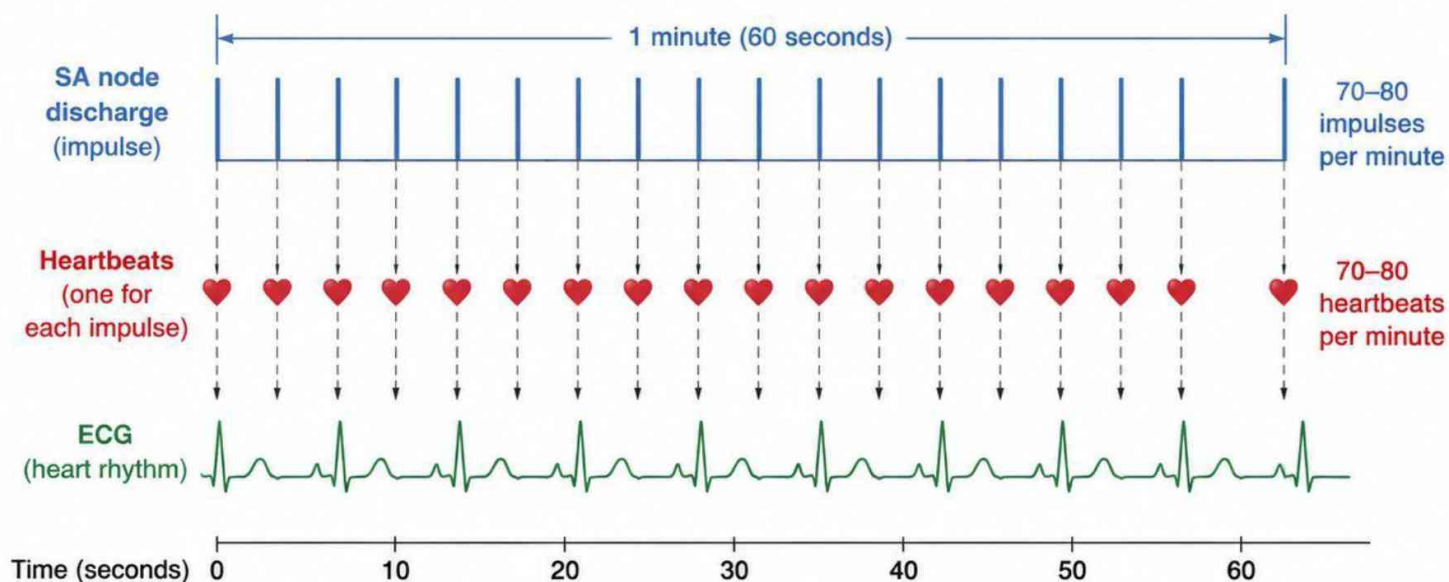
Dr. Doc Tutorial

SA NODE DISCHARGE 70–80 times per minute

- SA node (natural pacemaker) fires an electrical impulse 70–80 times per minute.
- Each impulse travels through the heart and produces one heartbeat.
- So, heart rate = SA node firing rate.



SA NODE FIRING AND HEARTBEATS



In 1 minute, SA node fires 70–80 times → produces 70–80 heartbeats

STANDARD BIPOLAR LIMB LEADS (I, II, III)

THE BASIS OF CLINICAL ELECTROCARDIOGRAPHY



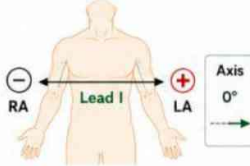
Bipolar leads record the electrical potential difference between two limbs.
They form **Einthoven's triangle** and are the foundation of ECG.

THE THREE BIPOLAR LEADS

1 LEAD I

- Negative (-) electrode on Right Arm (RA)
- Positive (+) electrode on Left Arm (LA)
- Records positive deflection when RA is negative relative to LA.

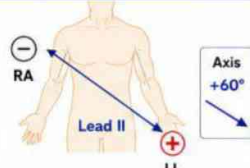
Axis: 0° (Horizontal)



2 LEAD II

- Negative (-) electrode on Right Arm (RA)
- Positive (+) electrode on Left Leg (LL)
- Records positive deflection when RA is negative relative to LL.

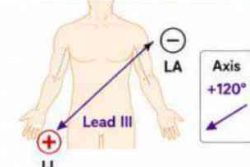
Axis: +60°



3 LEAD III

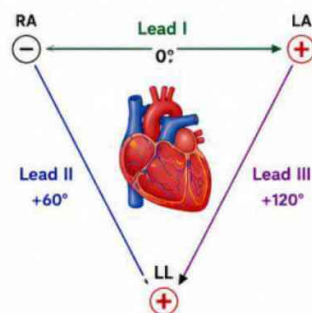
- Negative (-) electrode on Left Arm (LA)
- Positive (+) electrode on Left Leg (LL)
- Records positive deflection when LA is negative relative to LL.

Axis: +120°



EINTHOVEN'S TRIANGLE

The electrodes are placed on the limbs, forming an imaginary inverted triangle around the heart.



EINTHOVEN'S LAW

At any given instant, the algebraic sum of the potentials in any two leads equals the potential in the third lead.

$$\text{Lead I} + \text{Lead III} = \text{Lead II}$$

Example:

If Lead I = +0.5 mV and Lead III = +0.7 mV,
then Lead II = +1.2 mV

NORMAL ECG PATTERN IN EACH BIPOLAR LEAD

Lead I

Axis: 0°



- P wave: positive
- QRS complex: mostly positive
- T wave: positive

Lead II

Axis: +60°



- P wave: positive
- QRS complex: mostly positive
- T wave: positive

Lead III

Axis: +120°



- P wave: positive
- QRS complex: mostly positive
- T wave: positive

All three leads usually show similar waveforms in a healthy heart, but they "look" at the heart from different directions (axes).

CLINICAL SIGNIFICANCE

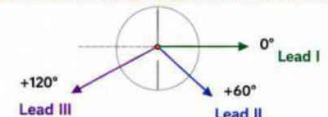
1. ARRHYTHMIA DIAGNOSIS

- Time relations between waves are similar in all three leads.
- Any lead can be used to diagnose heart rhythm issues (e.g., rate, regularity, heart block, ectopic beats).

2. DIAGNOSIS OF MYOCARDIAL DAMAGE (MYOPATHIES)

- Abnormalities (ST elevation/depression, Q waves, T changes) may appear in one lead and be minimal or absent in others.
- Recording from multiple leads helps detect the location and extent of damage or ischemia.

DIRECTIONS OF LEAD AXES (FRONTAL PLANE)



SUMMARY TABLE

Lead	Negative Electrode (-)	Positive Electrode (+)	Axis Direction	Normal QRS	Main View
I	Right Arm (RA)	Left Arm (LA)	0° (Horizontal)	Mostly Positive	Lateral wall
II	Right Arm (RA)	Left Leg (LL)	+60°	Mostly Positive	Inferior wall
III	Left Arm (LA)	Left Leg (LL)	+120°	Mostly Positive	Inferior wall

KEY POINTS

- Bipolar leads measure potential difference between two limbs.
- They form Einthoven's Triangle around the heart.
- Einthoven's Law: Lead I + Lead III = Lead II
- Lead II is usually the best for rhythm interpretation.
- Multiple leads are essential to detect myocardial abnormalities.



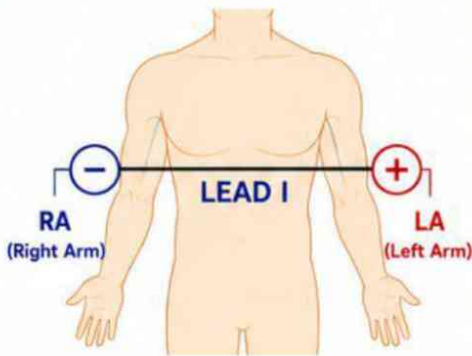
Remember: Electrical Activity of Heart → Body Fluids → Limbs (RA, LA, LL) → Electrodes → ECG Machine → ECG Tracing

LEAD I (STANDARD BIPOLAR LIMB LEAD)

Lead I is a bipolar lead that records the electrical potential difference between the **Left Arm (positive)** and **Right Arm (negative)**.

Dr. Doc

1. ELECTRODE ARRANGEMENT

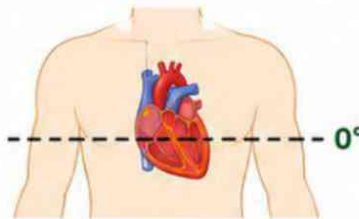


- Negative electrode (-) : Right Arm (RA)
- Positive electrode (+) : Left Arm (LA)

2. AXIS OF LEAD I

The electrical axis runs horizontally from right to left.

RA (-) → LA (+)
Axis = 0°



Lead I views the heart from the right side toward the left side (horizontal plane).

3. WHAT DOES LEAD I MEASURE?

Lead I compares:

LA - RA

It records whether the electrical activity is moving toward the **Left Arm**.

4. WHEN DOES LEAD I SHOW POSITIVE DEFLECTION?

If the depolarization wave travels toward the positive electrode (Left Arm):

RA (-) → LA (+)

The ECG tracing moves upward.

Movement toward Left Arm = Positive wave (upward)

5. TYPICAL APPEARANCE (NORMAL)

In a normal heart, the mean electrical activity is directed toward the left side (toward LA). Therefore, Lead I usually shows:

- ✓ Positive P wave
- ✓ Positive QRS complex
- ✓ Positive T wave



NORMAL LEAD I

- P wave : Positive
- QRS complex : Mostly positive
- T wave : Positive

6. SUMMARY TABLE

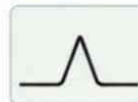
Lead	I
Negative Electrode (-)	Right Arm (RA)
Positive Electrode (+)	Left Arm (LA)
Axis	0° (Horizontal)
Direction	Right → Left
Measures	LA - RA
Positive Deflection	When impulse moves toward LA
Normal Waves	P ↑, QRS ↑, T ↑

7. UNDERSTANDING POSITIVE & NEGATIVE

- ✓ Impulse toward LA (+)

RA (-) → LA (+)

= Positive (upward deflection)



- ✗ Impulse toward RA (-)

LA (+) → RA (-)

= Negative (downward deflection)



- ⊕ Impulse perpendicular to axis

= Small or biphasic deflection



8. CLINICAL IMPORTANCE



Helps determine the mean electrical axis of the heart.



Useful in detecting left axis deviation or right axis deviation.



Helps assess left-sided cardiac electrical activity.



Abnormalities in Lead I may indicate ventricular hypertrophy, conduction defects or ischemia.



KEY POINT:

Lead I = LA - RA, Axis 0°, looks from Right to Left.

Normal ECG in Lead I shows **positive P wave, positive QRS** and **positive T wave**.

Dr. Doc Tutorial



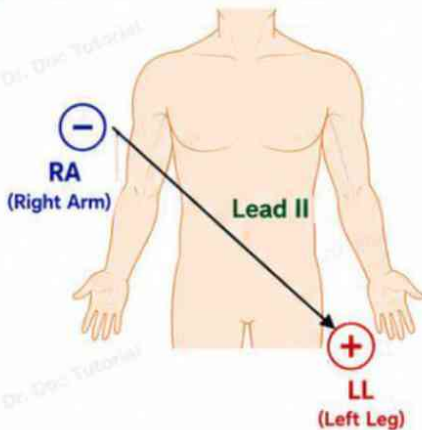
**Dr. Doc
Tutorial**

LEAD II (STANDARD BIPOLAR LIMB LEAD)

Lead II is a bipolar lead that records the electrical potential difference between the **Right Arm (negative)** and **Left Leg (positive)**.

**Dr. Doc
Tutorial**
— ECGal

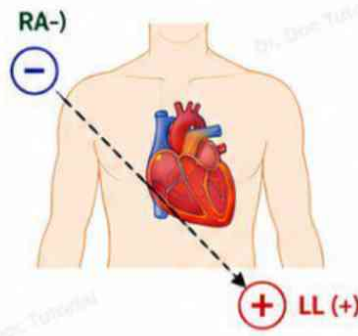
1. ELECTRODE ARRANGEMENT



- Negative electrode (-) : Right Arm (RA)
- Positive electrode (+) : Left Leg (LL)

2. AXIS OF LEAD II

The electrical axis runs from the **Right Arm** toward the **Left Leg**.



Axis = +60°

Lead II views the heart from the Right shoulder → Left leg direction.

3. WHAT DOES LEAD II MEASURE?

Lead II compares:

LL - RA

It records whether the electrical activity is moving toward or away from the **Left Leg** (positive electrode).

4. WHEN DOES LEAD II SHOW POSITIVE DEFLECTION?

If the depolarization wave travels toward the positive electrode (Left Leg):

RA (-) → LL (+)

The ECG tracing moves upward.

Movement toward Left Leg = Positive wave (upward)

5. TYPICAL APPEARANCE (NORMAL)

In a normal heart, because Lead II axis (+60°) is almost parallel to the normal direction of ventricular depolarization (+59°), it usually shows large positive waves.

- ✓ Positive P wave
- ✓ Large positive QRS complex
- ✓ Positive T wave



(Most commonly used rhythm lead because it gives the clearest and largest waves.)

NORMAL LEAD II

- P wave : **Positive**
- QRS complex : Predominantly positive (tall R wave)
- T wave : **Positive**

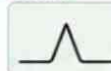
6. LEAD II - KEY POINTS

Lead	II
Negative Electrode (-)	Right Arm (RA)
Positive Electrode (+)	Left Leg (LL)
Axis	+60°
Direction	Right shoulder → Left leg
Measures	LL - RA
Why waves are large?	Axis (+60°) ≈ Normal cardiac vector (+59°)
Used for	Rhythm analysis, AV blocks, inferior wall MI

7. POSITIVE vs NEGATIVE DEFLECTION

- ✓ Impulse toward Left Leg (positive electrode)

RA (-) → LL (+)
= Positive deflection (upward)



- ✗ Impulse away from Left Leg (toward Right Arm)

RA (-) ← LL (+)
= Negative deflection (downward)



- ✗ Impulse perpendicular to axis

= Small or biphasic deflection



8. CLINICAL SIGNIFICANCE

- Best lead for rhythm analysis (sinus rhythm, arrhythmias).
- Excellent for detecting AV blocks (P wave, PR interval, QRS).
- Useful in diagnosing inferior wall myocardial infarction.
- P waves are clear → helps in detecting atrial abnormalities.

KEY TAKE-HOME MESSAGE

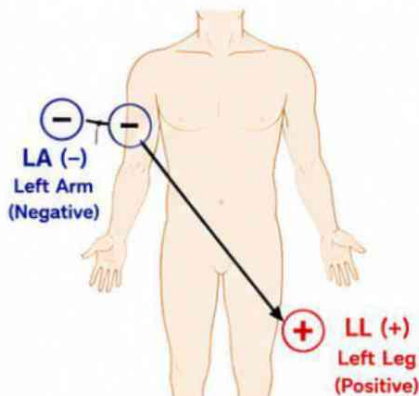
Lead II (RA - to LL +, axis +60°) is the most important bipolar limb lead. It produces the **clearest and largest P, QRS and T waves** because its axis is almost parallel to the heart's normal electrical activity.

Dr. Doc Tutorial — ECGal

LEAD III (STANDARD BIPOLAR LIMB LEAD)

Lead III is a bipolar lead that records the electrical potential difference between the **Left Arm (negative)** and **Left Leg (positive)**.

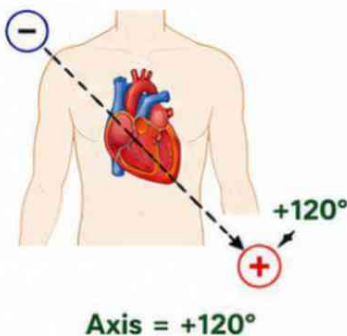
1. ELECTRODE ARRANGEMENT



- Negative electrode (-) : Left Arm (LA)
- Positive electrode (+) : Left Leg (LL)

2. AXIS OF LEAD III

The electrical axis runs from the **Left Arm** toward the **Left Leg**.



Lead III views the heart from the left side downward toward the left leg (inferior-left direction).

3. WHAT DOES LEAD III MEASURE?

Lead III compares:

$$LL - LA$$

It records whether the electrical activity is moving toward or away from the **Left Leg** (positive electrode).

4. WHEN DOES LEAD III SHOW POSITIVE DEFLECTION?

If the depolarization wave travels toward the positive electrode (Left Leg):

$$LA (-) \longrightarrow LL (+)$$

The ECG tracing moves upward.

Movement toward **Left Leg**
= **Positive wave (upward)**

5. TYPICAL APPEARANCE (NORMAL)

In a normal heart, the mean electrical activity is directed downward and to the left ($\approx +59^\circ$). Therefore, Lead III usually shows positive waves.

- ✓ Positive P wave
- ✓ Usually positive QRS complex (smaller than Lead II)
- ✓ Positive T wave



NORMAL LEAD III

- P wave : **Positive**
- QRS complex : Usually positive (may be smaller or biphasic)
- T wave : **Positive**

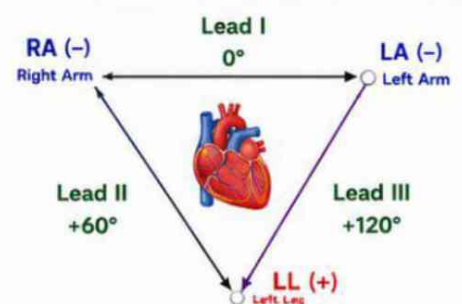
6. SUMMARY TABLE

Lead	III
Negative Electrode (-)	Left Arm (LA)
Positive Electrode (+)	Left Leg (LL)
Axis	+120°
Direction	Left Arm → Left Leg (Inferior-left)
Measures	LL - LA
Positive Deflection	When impulse moves toward LL
Normal Waves	P ↑, QRS ↑, T ↑

7. CLINICAL SIGNIFICANCE

- ♥ Part of Einthoven's Triangle (with Leads I and II).
- ⚖ Essential for Einthoven's Law: Lead I + Lead III = Lead II
- ♥ Helps determine the mean electrical axis of the heart.
- ♥ ST elevation in Lead III (with II and aVF) may indicate inferior wall myocardial infarction.
- 🫁 Useful in right ventricular hypertrophy and right axis deviation.

8. LEAD III IN EINTHOVEN'S TRIANGLE



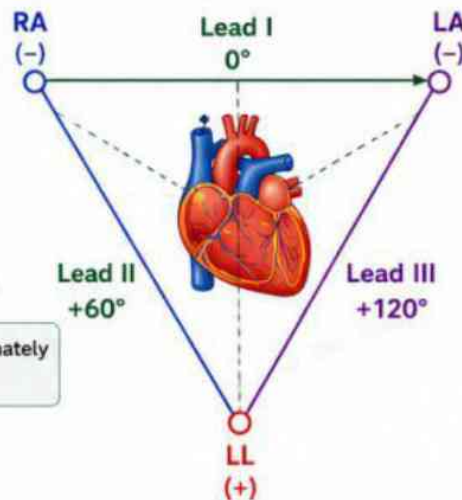
KEY POINT

Lead III (LA - LL, axis +120°) looks from Left Arm toward Left Leg. It usually shows **positive P, QRS and T waves**, but the amplitudes are smaller than in Lead II because its axis is less aligned with the normal ventricular depolarization vector.

EINTHOVEN'S TRIANGLE & EINTHOVEN'S LAW

1. EINTHOVEN'S TRIANGLE – DEFINITION

- An imaginary equilateral triangle formed by:
 - Right arm electrode (RA)
 - Left arm electrode (LA)
 - Left leg electrode (LL)

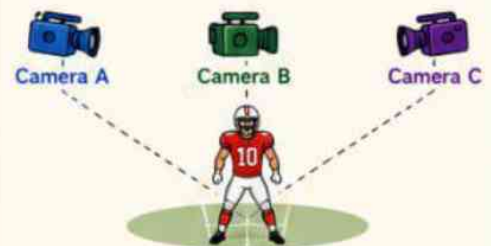


2. WHY IS EINTHOVEN'S TRIANGLE IMPORTANT?

- It provides standardized viewpoints for observing cardiac electrical activity.
- Each lead sees the same electrical event from a different direction.

ANALOGY

Imagine three cameras observing a football player.



The player is the same.

Each camera records a slightly different view.

Similarly, Lead I, II, and III view the same heartbeat from different angles.

3. EINTHOVEN'S LAW – STATEMENT

At any instant:

$$\text{Lead I} + \text{Lead III} = \text{Lead II}$$

This relationship always exists mathematically.

4. WHY DOES EINTHOVEN'S LAW WORK?

Recall:

$$\text{Lead I} = \text{LA} - \text{RA}$$

$$\text{Lead III} = \text{LL} - \text{LA}$$

Adding:

$$\begin{aligned} &(\text{LA} - \text{RA}) + (\text{LL} - \text{LA}) \\ &= \cancel{\text{LA}} - \text{RA} + \text{LL} - \cancel{\text{LA}} \\ &= \text{LL} - \text{RA} \\ &= \text{Lead II} \end{aligned}$$

LA terms cancel.

5. EXAMPLE

Suppose:

$$\text{Lead I} = +0.5 \text{ mV}$$

$$\text{Lead III} = +0.7 \text{ mV}$$

Then:

$$\begin{aligned} \text{Lead II} &= 0.5 + 0.7 \\ &= 1.2 \text{ mV} \end{aligned}$$

$$\begin{aligned} \text{Lead I} &= +0.5 \text{ mV} \\ \text{Lead III} &= +0.7 \text{ mV} \\ \hline \text{Lead II} &= +1.2 \text{ mV} \end{aligned}$$

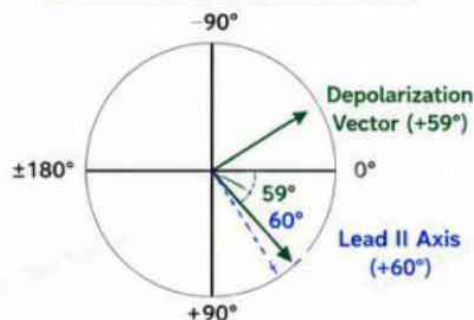
6. WHY IS LEAD II USUALLY THE LARGEST?

- The normal ventricular depolarization vector points approximately $+59^\circ$.
- Lead II lies at $+60^\circ$.
- These directions are almost identical.
- Therefore:

Depolarization vector
($+59^\circ$)
||
Lead II axis ($+60^\circ$)

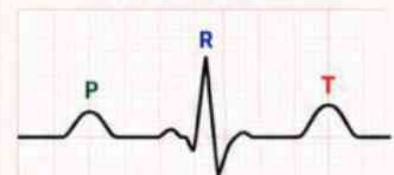
Nearly perfect alignment produces
maximal positive deflection.

FRONTAL PLANE (VIEW FROM FRONT)



RESULT

Lead II is almost perfectly aligned with the normal depolarization vector, so it records the largest positive P wave, QRS complex and T wave among the three bipolar leads.



KEY TAKE-HOME MESSAGE

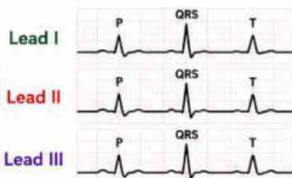
Einthoven's Triangle provides three standardized viewpoints (Leads I, II, III) around the heart, and Einthoven's Law ($\text{Lead I} + \text{Lead III} = \text{Lead II}$) is a fundamental mathematical relationship that always holds true.



CLINICAL SIGNIFICANCE OF BIPOLAR LIMB LEADS

1. DIAGNOSIS OF ARRHYTHMIAS

All three bipolar limb leads (I, II, III) show P waves, QRS complexes, and T waves at nearly identical times.



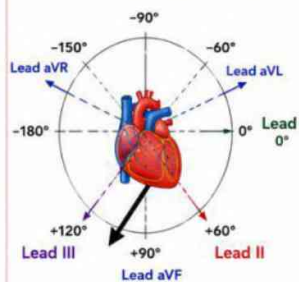
Therefore any lead can often identify:

- ✓ Sinus rhythm
- ✓ Tachycardia
- ✓ Bradycardia
- ✓ AV block

★ However, Lead II is usually preferred because waveforms are the largest and easiest to interpret.

2. DETERMINATION OF ELECTRICAL AXIS

Comparing Leads I, II, and III helps determine the Mean Electrical Axis (MEA) of ventricular depolarization.



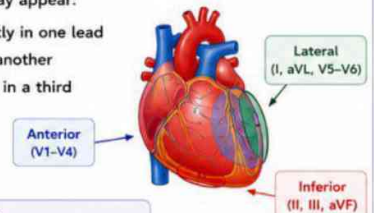
Axis deviation may indicate:

- ✓ Left ventricular hypertrophy (LVH)
- ✓ Right ventricular hypertrophy (RVH)
- ✓ Bundle branch block
- ✓ Fascicular block

3. DETECTION OF MYOCARDIAL INJURY

Different leads view different regions of the heart. Damage may appear:

- Prominently in one lead
- Mildly in another
- Not at all in a third



Examples:

- Inferior myocardial infarction
- Lateral myocardial infarction
- Conduction defects

★ Thus, multiple leads must be examined together for accurate diagnosis.

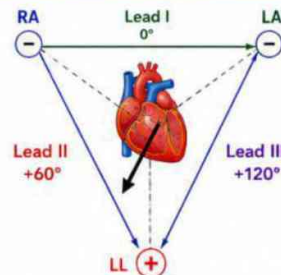
4. WHY DO ALL LEADS NOT LOOK IDENTICAL?

Because electrical vectors project differently onto each lead axis.

EXAMPLE:

A depolarization vector may point:

- ✓ Mostly toward Lead II
- ✓ Partly toward Lead I
- ✓ Minimally toward Lead III



Since the main depolarization vector is directed down and left ($\approx +59^\circ$), it is:

- ✓ Almost parallel to Lead II axis $\Rightarrow$ Large positive deflection
- ✓ Partly toward Lead I axis $\Rightarrow$ Moderate positive deflection
- ✓ Less toward Lead III axis $\Rightarrow$ Small positive deflection

RESULT: DIFFERENT AMPLITUDES

Same electrical event, different views (amplitudes).



KEY TAKE-HOME MESSAGE:

Bipolar limb leads provide multiple perspectives of the same cardiac electrical activity. They are essential for rhythm diagnosis, axis determination, and detection of myocardial injury.

AUGMENTED UNIPOLAR LIMB LEADS

(aVR, aVL and aVF)

DEFINITION

The augmented unipolar limb leads (aVR, aVL, and aVF) are ECG leads that record the electrical activity of the heart from one positive (exploring) electrode relative to an indifferent reference electrode that has nearly zero electrical potential.

Unlike the standard bipolar leads (I, II, III), which compare the voltage difference between two active electrodes, augmented leads observe the heart's electrical activity from a single viewpoint.

ONE VIEWPOINT (A CAMERA ANALOGY)



Watching the heart from one location

WHY ARE THEY CALLED "UNIPOLAR"?

BIPOLAR LEAD

Measures voltage between two active electrodes.

$$\text{Lead I} = \text{LA} - \text{RA}$$

$$\text{Lead II} = \text{LL} - \text{RA}$$

$$\text{Lead III} = \text{LL} - \text{LA}$$

Both electrodes actively contribute to the recording.

UNIPOLAR LEAD

Uses:

- One positive exploring electrode
- One indifferent (reference) electrode

Thus the ECG records electrical activity from the perspective of one limb.

Think of it as:
A camera viewing the heart from one location.

WHY ARE THEY CALLED "AUGMENTED"?

Originally, unipolar limb leads produced small voltage signals.

To make them easier to interpret, the ECG machine amplifies the recorded voltage by approximately 50%.

Hence:

$$a = \text{augmented}$$

$$V = \text{voltage}$$

aVR = Augmented Voltage Right arm

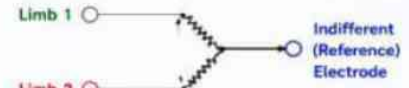
aVL = Augmented Voltage Left arm

aVF = Augmented Voltage Foot (left leg)

THE RECORDING MECHANISM

STEP 1: Create an Indifferent Electrode

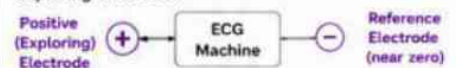
The ECG machine electrically combines two limbs using resistors.



The average electrical potential of these two limbs serves as the reference electrode. Because this averaged potential is near zero, it functions as an indifferent electrode.

STEP 2: Connect the Third Limb as Positive

The remaining limb becomes the positive exploring electrode.



The ECG then measures:

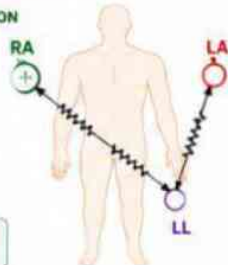
$$\text{Positive Electrode} - \text{Reference Electrode}$$

THE THREE AUGMENTED UNIPOLAR LIMB LEADS

1. LEAD aVR

ELECTRODE CONFIGURATION

- Positive electrode: Right Arm (RA)
- Reference electrode: Average of Left Arm and Left Leg



Mathematically:

$$aVR = RA - \frac{(LA + LL)}{2}$$

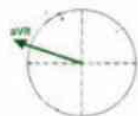
DIRECTION OF VIEW

Lead aVR views the heart from the right shoulder. It observes electrical activity moving toward or away from the right arm.

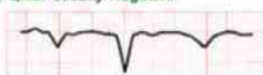


AXIS

-150°
(or +210°)



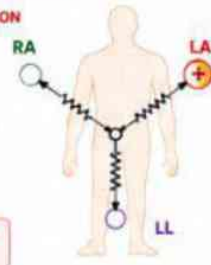
NORMAL QRS: Usually Negative



2. LEAD aVL

ELECTRODE CONFIGURATION

- Positive electrode: Left Arm (LA)
- Reference electrode: Average of Right Arm and Left Leg



Mathematically:

$$aVL = LA - \frac{(RA + LL)}{2}$$

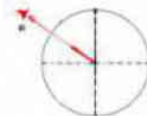
DIRECTION OF VIEW

Lead aVL views the heart from the left shoulder. It primarily examines the lateral aspect of the left ventricle.



AXIS

-30°



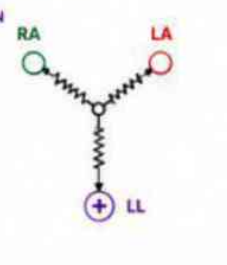
NORMAL QRS: Usually Positive



3. LEAD aVF

ELECTRODE CONFIGURATION

- Positive electrode: Left Leg (Foot)
- Reference electrode: Average of Right Arm and Left Arm

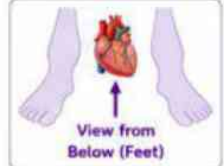


Mathematically:

$$aVF = LL - \frac{(RA + LA)}{2}$$

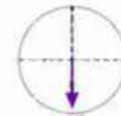
DIRECTION OF VIEW

Lead aVF views the heart from below (inferior aspect). It primarily examines the inferior surface of the heart.



AXIS

+90°



NORMAL QRS: Usually Positive

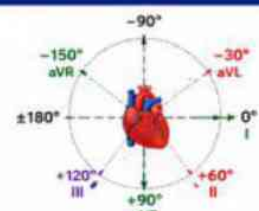


SUMMARY TABLE

Lead	Positive Electrode	Reference (Indifferent) Electrode	Axis	Direction	Normal QRS
aVR	Right Arm (RA)	Average of LA + LL	-150° (or +210°)	Right Shoulder	Usually Negative
aVL	Left Arm (LA)	Average of RA + LL	-30°	Left Shoulder	Usually Positive
aVF	Left Leg (LL)	Average of RA + LA	+90°	Inferior (Feet)	Usually Positive

KEY POINTS

- ★ Augmented leads are unipolar (one exploring electrode).
- ★ The reference electrode is the average of the other two limbs.
- ★ aVR is usually negative because the normal depolarization vector (+59°) points away from the right arm (-150°).
- ★ aVL and aVF are usually positive.
- ★ These leads are essential for axis determination and diagnosing various cardiac conditions.



ONE-LINE TAKE HOME MESSAGE

Augmented unipolar limb leads (aVR, aVL, aVF) use one positive electrode referenced to an averaged indifferent electrode to provide three additional viewpoints crucial for axis determination and detection of cardiac abnormalities.



Dr. Doc
Tutorial



PRECORDIAL (CHEST) LEADS (V₁-V₆)



Dr. Doc
Tutorial

DEFINITION

Precordial (chest) leads are six unipolar leads (V₁-V₆) that record the electrical activity of the heart from the **horizontal (transverse) plane**.

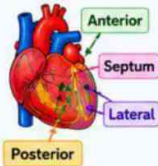
WHY ARE CHEST LEADS IMPORTANT?

✓ Limb leads show:

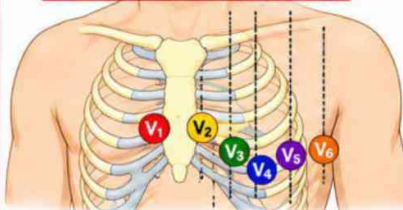
Right ↔ Left
Up ↔ Down

But they give limited information about:

- Anterior wall
- Posterior wall
- Septum
- Lateral wall



1. ELECTRODE POSITIONS (V₁-V₆)



V₁ 4th ICS, Right parasternal (right sternal border)

V₂ 4th ICS, Left parasternal (left sternal border)

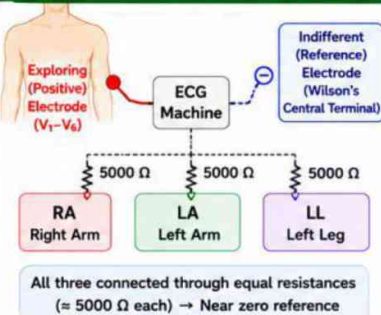
V₃ Midway between V₂ and V₄

V₄ 5th ICS, Midclavicular line

V₅ Anterior axillary line (same level as V₄)

V₆ Midaxillary line (same level as V₄ & V₅)

2. HOW THEY ARE CONNECTED (UNIPOLAR)



SIMPLE ANALOGY

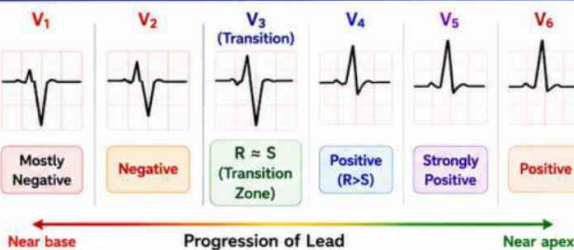
Limb leads = photos from above (Up-Down, Right-Left)



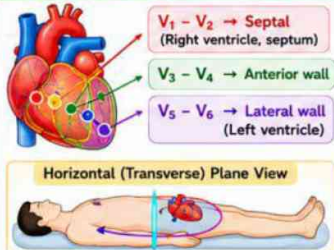
Chest leads = six cameras around the front and left side (see septum, anterior & lateral walls clearly)



3. NORMAL QRS PROGRESSION (V₁ → V₆)

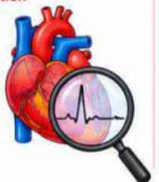


4. AREAS VIEWED BY EACH LEAD



5. CLINICAL IMPORTANCE

- ✓ Detects **anterior, lateral, and septal** infarction
- ✓ Diagnoses bundle branch blocks
- ✓ Identifies **ventricular hypertrophy**
- ✓ Detects small conduction abnormalities (sensitive because electrodes are close to the heart)



KEY TAKE-HOME MESSAGE:

Precordial leads (V₁-V₆) are **unipolar** chest leads that use Wilson's Central Terminal (RA + LA + LL) as the reference. They record activity in the **horizontal plane** and show normal progression from **negative** in V₁-V₂ to **positive** in V₄-V₆.

Dr. Doc Tutorial

PRECORDIAL (CHEST) LEADS (V₁-V₆)

DETAILED EXPLANATION

1. DEFINITION

The precordial leads (also called chest leads) are six unipolar ECG leads designated:

V₁, V₂, V₃, V₄,
V₅, V₆

They record the heart's electrical activity from the horizontal (transverse) plane, whereas the limb leads observe electrical activity mainly in the frontal plane.

Chest leads = Horizontal view
Limb leads = Frontal view

2. WHY ARE CHEST LEADS IMPORTANT?

Limb leads tell us whether electrical activity is moving:

Right ↔ Left Up ↔ Down

However, they provide limited information about:

- Anterior wall
- Posterior wall
- Septum
- Lateral wall

The chest leads solve this problem by viewing the heart from the front and left side.

SIMPLE ANALOGY

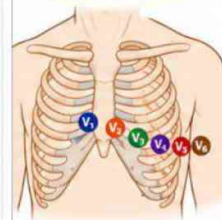
Imagine taking photographs of a building. You already have photographs from above (limb leads). To see the front wall clearly, you need cameras placed around the building.



The precordial leads act as six cameras positioned across the chest. Each lead views the heart from a different horizontal angle.

3. ELECTRODE ARRANGEMENT AND CONNECTION

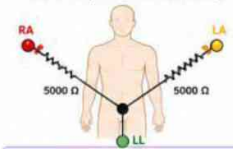
A. POSITIVE (EXPLORING) ELECTRODES: SIX CHEST POSITIONS



- V₁ 4th intercostal space Right sternal border
- V₂ 4th intercostal space Left sternal border
- V₃ Midway between V₂ and V₄
- V₄ 5th intercostal space Midclavicular line
- V₅ Anterior axillary line Same horizontal level as V₄
- V₆ Midaxillary line Same horizontal level as V₄ and V₅

B. NEGATIVE (INDIFFERENT) ELECTRODE: WILSON'S CENTRAL TERMINAL

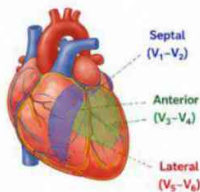
Right Arm (RA), Left Arm (LA), and Left Leg (LL) are connected through equal resistances (≈ 5000 Ω each). This combined point acts as a near-zero reference (indifferent electrode).



ECG measures: Positive Chest Electrode - Wilson's Central Terminal

4. AREAS VIEWED BY EACH LEAD

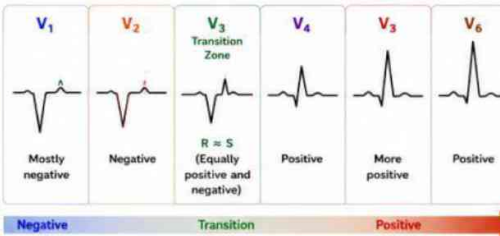
Lead	Region Viewed
V ₁	Right ventricle & septum
V ₂	Septum
V ₃	Anterior septal region
V ₄	Anterior wall
V ₅	Left ventricular lateral wall
V ₆	Left ventricular lateral wall



Quick Memory: V₁-V₂ = Septal | V₃-V₄ = Anterior | V₅-V₆ = Lateral

5. NORMAL QRS PROGRESSION ACROSS THE CHEST

As we move from V₁ to V₆, the QRS complex changes gradually from mostly negative to mostly positive.



6. WHY DOES THE QRS PATTERN CHANGE?

V₁-V₂ (Near the Base)

Electrodes are near the base of the heart. During most of ventricular depolarization, the base is relatively electronegative while the apex is electropositive. Vector moves AWAY → Negative QRS.



V₃ (Mid Zone)

Electrodes are midway between the base and apex. Positive and negative forces are nearly equal → Transition zone.



V₄-V₆ (Near the Apex/Left Ventricle)

Electrodes are closer to the apex and left ventricle. The major depolarization vector moves TOWARD the electrodes → Positive QRS.



7. CLINICAL IMPORTANCE



Anterior Myocardial Infarction Affects V₃-V₄



Septal Infarction Affects V₁-V₂



Lateral Wall Infarction Affects V₅-V₆



Bundle Branch Blocks Alter QRS morphology especially in V₁ and V₆



Ventricular Hypertrophy RVH → Tall R in V₁ LVH → Tall R in V₅-V₆

8. KEY POINTS TO REMEMBER

- ✓ Precordial leads (V₁-V₆) record electrical activity in the horizontal plane.
- ✓ One positive chest electrode is used at six standard positions.
- ✓ The negative electrode is Wilson's Central Terminal (RA + LA + LL through equal resistances).
- ✓ They are highly sensitive to anterior, septal, and lateral cardiac abnormalities.
- ✓ Normal QRS progression: Negative (V₁-V₂) → Transition (V₃) → Positive (V₄-V₆).

SUMMARY

Chest leads provide a horizontal view of the heart and are essential for detecting localized myocardial and conduction abnormalities.

AXIS DEVIATION (MEAN QRS AXIS DEVIATION)

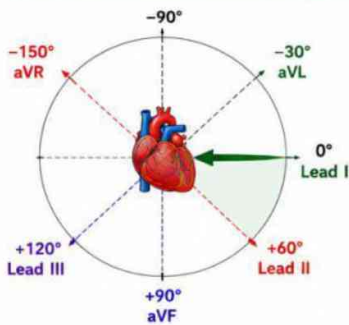
Normal Mean Electrical Axis: $+20^{\circ}$ to $+100^{\circ}$ (average $+59^{\circ}$)
Directed downward and leftward toward the apex.

AXES OF FRONTAL PLANE LEADS

Lead I	=	0°
Lead II	=	$+60^{\circ}$
Lead III	=	$+120^{\circ}$
aVR	=	-150° ($+210^{\circ}$)
aVL	=	-30°
aVF	=	$+90^{\circ}$

LEFT AXIS DEVIATION (LAD)

Axis shifts leftward to -30° or more negative.
Axis usually $< -30^{\circ}$



COMMON CAUSES

- Pathological Causes**
 - Left Ventricular Hypertrophy (LVH): Increased muscle mass of LV → greater electrical force toward left. Causes: Hypertension, Aortic stenosis, Aortic regurgitation
 - Left Bundle Branch Block (LBBB): Right ventricle depolarizes first; strong late vector toward LV → axis around -50° . QRS duration > 0.12 sec
- Physical / Physiological Factors**
 - Lying down (supine position): abdominal contents push diaphragm upward → axis shifts leftward.
 - Deep expiration → axis shifts leftward.
 - Obesity → diaphragm pushed upward chronically → permanent leftward shift.

TYPICAL AXIS RANGE
 -30° to -90°
(Leftward)
Most common
around -45° to -60°

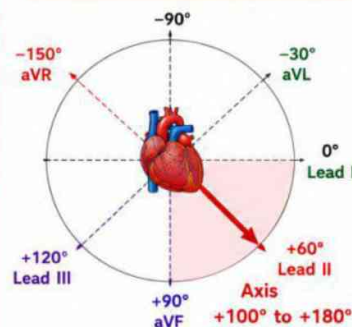
ECG PATTERN (HOW TO RECOGNIZE)



- KEY POINTS**
- Axis points more toward the left shoulder.
 - Lead I positive, aVF negative.
 - Common causes: LVH, LBBB, obesity, supine position, expiration.

RIGHT AXIS DEVIATION (RAD)

Axis shifts rightward to $+100^{\circ}$ or more.
Axis usually $> +100^{\circ}$



COMMON CAUSES

- Pathological Causes**
 - Right Ventricular Hypertrophy (RVH): Increased mass in RV → greater electrical force toward right. Causes: Pulmonary stenosis, Tetralogy of Fallot, VSD, Pulmonary hypertension
 - Right Bundle Branch Block (RBBB): Left ventricle depolarizes first; late RV activation → strong rightward vector. QRS duration > 0.12 sec
- Physical / Physiological Factors**
 - Standing up → axis shifts rightward.
 - Deep inspiration → axis shifts rightward.
 - Tall, lanky individuals → heart hangs vertically → natural rightward axis.

TYPICAL AXIS RANGE
 $+100^{\circ}$ to $+180^{\circ}$
(Rightward)
Most common
around $+120^{\circ}$ to $+150^{\circ}$

ECG PATTERN (HOW TO RECOGNIZE)



- KEY POINTS**
- Axis points more toward the right shoulder.
 - Lead I negative, aVF positive.
 - Common causes: RVH, RBBB, standing position, inspiration, tall thin body build.

Remember: Determine axis by using Leads I and aVF first:

I (+) aVF (+) → Normal Axis ($+20^{\circ}$ to $+100^{\circ}$)

I (+) aVF (-) → Left Axis Deviation ($< -30^{\circ}$)

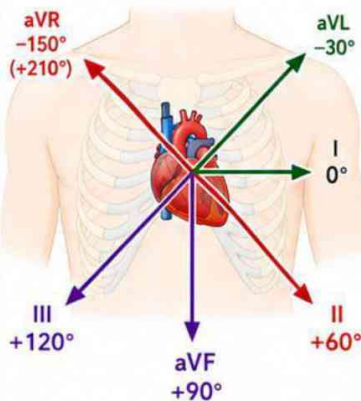
I (-) aVF (+) → Right Axis Deviation ($> +100^{\circ}$)

THE HEXAXIAL (HEXAGONAL) REFERENCE SYSTEM

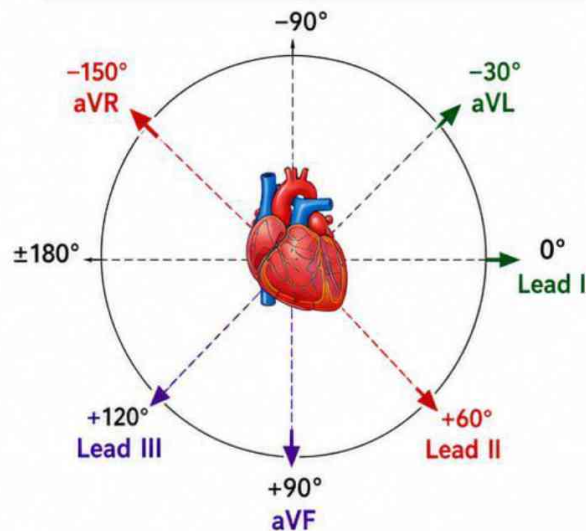
A diagrammatic tool used in electrocardiography to determine the mean electrical axis of the heart in the frontal plane.

WHAT IS IT?

It represents the axes of the six frontal plane leads as they radiate from the center of the heart.



AXES OF THE FRONTAL PLANE LEADS



LEAD AXES TABLE

Lead	Angle	Direction (from - to +)
Standard Bipolar Leads		
Lead I	0°	Right Arm (-) → Left Arm (+)
Lead II	+60°	Right Arm (-) → Left Leg (+)
Lead III	+120°	Left Arm (-) → Left Leg (+)
Augmented Unipolar Leads		
aVL	-30°	Average of (RA + LL) (-) → Left Arm (+)
aVF	+90°	Average of (RA + LA) (-) → Left Leg (+)
aVR	+210° (-150°)	Average of (LA + LL) (-) → Right Arm (+)

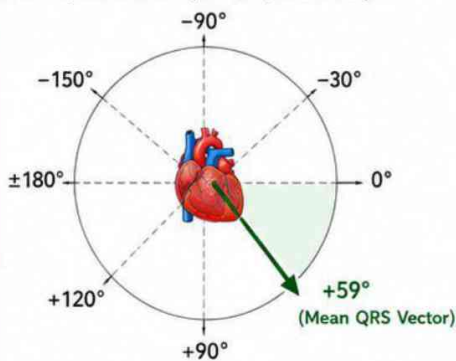
DETERMINING THE MEAN ELECTRICAL AXIS (MEA)

The MEA is the preponderant direction of the electrical potential during ventricular depolarization (mean QRS vector).

NORMAL MEAN ELECTRICAL AXIS

Typically directed toward the apex at approximately **+59°**.

Normal Range
+20° to +100°

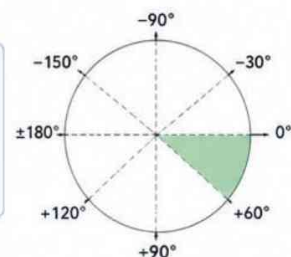


HOW TO USE THE HEXAXIAL SYSTEM

1. Examine the QRS complex in Leads I and aVF.
2. Determine if the QRS is predominantly positive (+), negative (-), or isoelectric (±) in each lead.
3. Locate the region on the hexaxial diagram where the MEA must lie based on those findings.
4. Refine the axis using Leads II and III (if needed).

EXAMPLE

Lead I: Positive
| aVF: Positive
→ Axis lies between 0° and +90°



Further analysis with Leads II and III places the MEA at approximately **+59° (normal)**



KEY POINTS

- The hexaxial system includes six frontal plane leads: I, II, III, aVR, aVL, aVF.
- Each lead has a fixed axis.
- The MEA is the average direction of ventricular depolarization.
- Normal axis: +20° to +100° (mean ≈ +59°).



**Dr. Doc
Tutorial**

